Package 'CJQR'

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Type Package

Title Copula-Based Joint Quantile Regression

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Description Fits bivariate additive location models with enriched asymmetric-Laplace margins and Gaussian, Clayton, Frank, Gumbel or Student-t copulas. Adaptive anchor selection, fixed-degree marginal shapes, constant dependence and margin-specific covariate bases provide conditional quantile predictions from one fitted distribution. Auxiliary-bootstrap pointwise quantile intervals use a paired-subject least-squares and residual-quantile bootstrap without refitting the joint likelihood. These intervals require independent sampling and common-error location assumptions; they do not provide joint parameter covariance. Experimental extensions supply auxiliary-bootstrap intervals and tests for marginal parametric effects and whole smooth components, with conditional joint-information complexity diagnostics. Student-t copula degrees of freedom are supplied by the user and held fixed.

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Author Guanjie Lyu [aut, cre]

Maintainer Guanjie Lyu <guanjielyu@gmail.com>

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`CJQR-package` *CJQR: Copula-Based Joint Quantile Regression*

Description

Joint models for two continuous responses, with additive covariate effects and a copula. The primary interface uses enriched asymmetric-Laplace (EAL) margins and supplies conditional quantiles from a single fitted distribution.

Details

The package name `CJQR` is case-sensitive. Public function names use lowercase `cjqr`. Start with `cjqr_example_data` and `cjqr_eal`, then inspect `cjqr_fit_diagnostics`.

The EAL model allows different marginal bases, with separate coefficient vectors, fixed degree three, and constant marginal scales, shapes and copula dependence. Covariate effects are shared across quantile probabilities within a margin. Supported copulas are Gaussian, Clayton, Frank, Gumbel and Student-t. Student-t copula degrees of freedom must be supplied and are held fixed; this does not specify Student-t margins.

An analysis workflow

1. Generate an illustration with `cjqr_example_data`, or prepare independent paired continuous responses and observed covariates.
2. Fit `cjqr_eal` with two location formulas, each including an intercept. Their predictors and smooth bases may differ. Use `cjqr_eal_control` for validated numerical controls.
3. Inspect the equation-wise `summary`, numerical diagnostics and smooth plots. Plots include auxiliary-bootstrap pointwise intervals by default; use `se = FALSE` for estimates alone. Convergence does not establish model adequacy. Use `residuals.cjqr_eal` for marginal diagnostic residuals and `cjqr_residual_diagnostics` for joint Rosenblatt diagnostics.
4. Use `predict` for complete quantiles on a supplied covariate grid.
5. If the auxiliary-bootstrap assumptions are defensible, explicitly call `cjqr_eal_ci` for pointwise intervals. Use `as.data.frame` on its result for a long-format table.

Uncertainty and limitations

The EAL intervals enclose an auxiliary subject-pairs least-squares and residual-quantile bootstrap interval and the unchanged joint estimate. They are not native joint standard errors, EAL-bootstrap percentile intervals, simultaneous bands or future-outcome prediction intervals. They require the common-error location, basis and independent-sampling assumptions described in `cjqr_eal_ci`. Smooth plot intervals use centered bootstrap smooth contributions, without residual-quantile offsets; see `cjqr_eal_methods`. Separate experimental auxiliary-bootstrap APIs provide marginal coefficient intervals and tests (`cjqr_eal_coef_ci`), whole-smooth tests (`cjqr_eal_smooth_test`), and a Kendall's tau interval (`cjqr_eal_tau_ci`). None supplies native joint covariance or theta-scale Wald inference. Numerical success and examples do not establish uniform statistical superiority or general coverage.

Installed guides

Use `system.file("notes", "user-guide.md", package = "CJQR")` to find the EAL user guide. The installed example at `system.file("examples", "quick_demo.R", package = "CJQR")` shows a reproducible workflow. Package installation does not require LaTeX.

Author(s)

Guanjie Lyu <guanjielyu@gmail.com>

See Also

`cjqr_eal`, `cjqr_eal_ci`, `cjqr_eal_control`, `cjqr_fit_diagnostics`, `cjqr_example_data`. Use `citation("CJQR")` for the software citation.

Examples

```
dat <- cjqr_example_data(n = 50, seed = 1)
str(dat)
cjqr_eal_control()
system.file("notes", "user-guide.md", package = "CJQR")
## Not run:
fit <- cjqr_eal(
  list(y1 ~ z + s(x, k = 6), y2 ~ z + s(x, k = 6)),
  data = cjqr_example_data(n = 200, seed = 1),
  copula = "gaussian", seed = 1
)
summary(fit)
cjqr_fit_diagnostics(fit)

## End(Not run)
```

`as.data.frame.cjqr_eal_ci` *Convert joint EAL quantile intervals to a tidy table*

Description

Converts the arrays or response-named lists of matrices returned by `cjqr_eal_ci` into a data frame, without changing estimates or bounds. Prediction-row identity, response labels, target probabilities and interval availability are retained. No standard errors are inferred from interval widths.

Usage

```
## S3 method for class 'cjqr_eal_ci'
as.data.frame(x, row.names = NULL,
  optional = FALSE, include_reference = FALSE, ...)
```

Arguments

x	An object returned by cjqr_eal_ci.
row.names	Optional row names for the resulting long-format table, following data.frame conventions. These are separate from original prediction-row labels, which are preserved in a column.
optional	Logical flag controlling automatic column-name checking, as for other as.data.frame methods. The standard column names already satisfy R naming conventions.
include_reference	Also include the auxiliary point estimate and percentile bounds? Defaults to FALSE.
...	No additional arguments are supported.

Details

Rows are ordered by original prediction-row index, then response, then the supplied order of probabilities within that response. A margin-specific request may contain a different number of probabilities for each response; the prob column records the actual level for that response, not a shared column index. Response names come from the prediction array or matrix-list names; if absent in an array, the labels margin1 and margin2 are used. Original row labels need not be unique and are not used as the output table's row names.

When the auxiliary bootstrap fails, unchanged joint estimates are still reported, with NA bounds and available = FALSE. The strict all-draw policy is retained: this conversion neither drops failed rows nor fills in bounds. An array with zero prediction rows produces a zero-row table with metadata. Malformed arrays or matrix lists, inconsistent dimensions or availability flags, and intervals that fail the stated containment rules are rejected.

The cjqr_interval attribute stores the confidence level, method, availability and failure message, bootstrap draw counts and applicability assumptions. These describe auxiliary-bootstrap pointwise intervals, not native joint-model standard errors or simultaneous bands. Attributes may be dropped by downstream data manipulation or text export; retain the original interval object for a complete audit trail.

Value

A data frame with columns row_index, response, prob, estimate, lower, upper and available. A row_label column is added when original prediction-row labels exist. With include_reference = TRUE, three further columns are returned: reference_estimate, reference_lower and reference_upper. These names and include_reference are retained for compatibility and refer to the auxiliary estimator and its bounds.

See Also

cjqr_eal_ci, cjqr_example_data

Examples

```
## Not run:
dat <- cjqr_example_data(n = 200, seed = 1)
fit <- cjqr_eal(list(y1 ~ z + s(x, k = 6), y2 ~ z + s(x, k = 6)),
                data = dat, anchors = 0.5, seed = 1)
prediction_data <- data.frame(x = seq(-1, 1, length.out = 51), z = 0)
ci <- cjqr_eal_ci(fit, newdata = prediction_data)
interval_table <- as.data.frame(ci)
head(interval_table)
attr(interval_table, "cjqr_interval")

# Different levels in the two margins retain their own probability labels.
ci_margins <- cjqr_eal_ci(fit, newdata = prediction_data,
                          probs = list(y1 = c(0.1, 0.5), y2 = 0.9))
as.data.frame(ci_margins, include_reference = TRUE)

## End(Not run)
```

`cjqr_eal` *Fit an adaptive-anchor enriched asymmetric-Laplace joint location model*

Description

Fits a bivariate additive location model using degree-three enriched asymmetric-Laplace (EAL) marginal distributions and constant copula dependence. The fitting strategy uses Gaussian-GAM REML smoothing penalties, anchor-specific initial values, multistart optimization, and strict convergence checks.

Usage

```
cjqr_eal(formulas, data,
  copula = c("gaussian", "clayton", "frank", "gumbel", "student_t"),
  anchors = c(0.25, 0.5, 0.75), na.action = stats::na.fail,
  seed = 1L, control = list(), extra_control = list(), df = NULL,
  progress = FALSE, probs = c(0.1, 0.25, 0.5, 0.75, 0.9))

## S3 method for class 'cjqr_eal'
predict(object, newdata = NULL,
  probs = NULL,
  type = c("quantile", "location", "lpmatrix", "cdf"),
  interval = c("none", "confidence"), level = 0.95,
  B = 500L, seed = 1L, y = NULL, ...)

## S3 method for class 'cjqr_eal'
coef(object, ...)
## S3 method for class 'cjqr_eal'
fitted(object, probs = NULL, ...)
## S3 method for class 'cjqr_eal'
nobs(object, ...)
## S3 method for class 'cjqr_eal'
formula(x, ...)
```

```
## S3 method for class 'cjqr_eal'
model.frame(formula, ...)
## S3 method for class 'cjqr_eal'
vcov(object, ...)
## S3 method for class 'cjqr_eal'
confint(object, parm, level = 0.95, newdata = NULL,
  probs = NULL, B = 500L, seed = 1L, ...)
## S3 method for class 'cjqr_eal'
logLik(object, ...)
```

Arguments

formulas	A list of two formulas with distinct bare numeric response names and an explicit intercept in each equation. Right-hand sides may contain different covariates, smooth terms and basis dimensions. Each margin has its own regression coefficients. Neither response may appear as a predictor in either equation. An optional third formula must be `~ 1`, denoting constant dependence. Offsets and covariate-dependent dependence equations are not supported.
data	A data frame containing complete paired continuous responses and the covariates in both formulas. At least 30 complete paired observations are required. No observation-weight argument is supported.
copula	Gaussian, Clayton, Frank, Gumbel, or Student-t copula. Gaussian, Frank, and Student-t support positive and negative dependence; Clayton and Gumbel support positive dependence. The aliases `"t"` and `"student"` select `"student_t"`. This argument does not select a marginal distribution.
df	Required finite positive degrees of freedom for a Student-t copula. The value is fixed, not estimated. It must be `NULL` for other copula families. This is not a marginal distribution parameter.
anchors	Candidate probabilities strictly between zero and one for the common EAL anchor. The default candidate set is `c(0.25, 0.5, 0.75)`. A single probability fixes the anchor. The anchor is a distribution parameterization choice, not a requested prediction probability and not Kendall's tau.
na.action	Either `stats::na.fail` or `stats::na.omit`. The default `stats::na.fail` rejects incomplete observations rather than silently changing the paired analysis sample. Omission uses one common complete-case sample across both responses and the union of both formulas' covariates, and records the omitted row indices.
seed	A nonnegative integer local random seed for reproducible initialization or auxiliary bootstrap resampling. The caller's random-number stream is restored.
progress	Logical. If `TRUE`, emit messages before and after each anchor fit and report the selected anchor. The default is quiet. This option does not change optimization, random seeds or acceptance rules.
control	A named list of numerical fitting controls. Supported names are `maxit`, `retry_maxit`, `continuation_maxit`, `recovery_maxit`, `factr`, `pgtol`, `strict_gradient_tolerance`, `theta_init`, `rho_bounds`, `theta_bounds`, `frank_bounds`, `gumbel_bounds`, `shape_bound`, and `trace`. The `rho_bounds` control applies only to Gaussian or Student-t; `theta_bounds` applies only to Clayton. Frank uses signed `frank_bounds`, by default `c(-30, 30)`. Gumbel uses `gumbel_bounds`, by default `c(1 + exp(-17), 30)`, with both endpoints greater than one. Bounds for an unrelated family are rejected. The strict gradient tolerance cannot exceed `1e-6`. Use `cjqr_eal_control()`

to validate overrides before fitting. The controls do not enable a different EAL degree, observation weights, covariate-dependent scales, or a covariate-dependent copula. Unsupported settings are rejected.

`extra_control` Reserved for future extensions. Only the empty list is currently supported; the validated multistart strategy cannot be disabled through this argument.

`object`, `x`, `formula`
A certified `cjqr_eal` fitted object. The argument named `formula` in `model.frame` is the fitted object, not a new model formula.

`newdata` A nonempty prediction data frame containing the union of both formulas' fitted covariates and their original factor levels, or `NULL` to use the retained analysis observations.

`probs` A nonempty numeric vector of distinct quantile probabilities strictly between zero and one, shared by both margins, or a list of two such vectors for margin-specific probabilities. Within-margin order is retained. A named list must use exactly the two fitted response names; entries are reordered to fitted response order. An unnamed list follows that order directly. For example, `list(y1 = 0.1, y2 = 0.9)` requests different levels for responses `y1` and `y2`, whereas `c(0.1, 0.9)` requests both levels for both margins. An explicit fitting argument stores the preferred prediction targets; it does not change the likelihood or anchor selection. If omitted when fitting, no preferred targets are stored. For `predict`, `fitted` and `confint`, `NULL` uses those stored targets, or `c(0.1, 0.25, 0.5, 0.75, 0.9)` when none are stored, including for older fitted objects.

`type` Return complete marginal quantiles, anchor-location predictors, the marginal design matrices, or the two marginal conditional CDF values. `"lpmatrix"` returns one matrix for identical right-hand sides, formula environments and represented bases, otherwise a response-named list of two matrices. The CDF option does not return a joint copula probability.

`interval` By default, no intervals are computed. For quantile predictions, `"confidence"` explicitly invokes `cjqr_eal_ci()`.

`level` Confidence level strictly between zero and one.

`B` Number of auxiliary subject-pairs bootstrap resamples; at least two, with default 500. This is not the number of joint-model refits.

`y` For CDF prediction, a finite numeric matrix with one row per prediction observation and columns in the fitted response order. If omitted, the observed responses or response columns of `newdata` are used when available.

`parm` Must be omitted. This `confint` method returns complete marginal quantile intervals. Use `cjqr_eal_coef_ci` for auxiliary-bootstrap marginal parametric coefficient intervals.

`...` Additional generic-method arguments. Prediction rejects unrecognized arguments. With `interval = "confidence"`, it also accepts `chunk_size`, `max_memory_mb`, and `keep_draws` for `cjqr_eal_ci()`. `confint` forwards arguments to `cjqr_eal_ci()`.

Details

For margin j, the model has the form

$$Y_j = \eta_j(x) + \sigma_j E_j,$$

where $\eta_j(x)$ is represented by margin j's additive basis, the scale σ_j is constant, and E_j has an EAL distribution with constant shape parameters. Both margins use a common selected anchor a, with $F_{E_j}(0) = a$. The polynomial degree is fixed at three; degree selection is not performed.

The two location formulas may differ. For example, one margin can include a coordinate-based bivariate smooth while the other contains only a univariate smooth. Basis construction, smoothing penalties and coefficient blocks remain margin-specific. A coordinate smooth represents a location effect, not a residual spatial-dependence model; it does not establish independent sampling for spatially correlated observations.

The complete conditional quantile is

$$q_{jp}(x) = \eta_j(x) + \sigma_j F_{E_j}^{-1}(p).$$

Consequently the model shares each margin's covariate effects across probabilities: quantile curves differ by constant intercept shifts. Only at the selected anchor is the marginal quantile equal to the location predictor. The same fitted distribution supplies all requested probabilities; separate responses or separate likelihood fits are not generated for each probability. The `probs` argument only selects reported quantiles, including different levels or numbers of levels for the two margins. It is independent of the common candidate `anchors` and never changes the fitted distribution. A two-element numeric `probs` vector is a shared prediction grid; use a list for margin-specific targets. To fit at a single fixed level, supply that probability through `anchors`, for example `anchors = 0.10`.

Gaussian `mgcv` fits select the baseline smoothing penalties. These penalties are held fixed during the joint optimization. Anchor-specific `qgam` fits supply starting values, not covariance estimates for the joint model. The fitted object records initialization and anchor-search diagnostics. Selection compares admissible candidates using the penalized likelihood objective and the strict numerical acceptance rules; an optimizer return code alone is not a convergence certificate. No simulated truth or comparison with another estimator is used to select a candidate.

Use `cjqr_fit_diagnostics()` to inspect anchor and initializer checks. An uncertified selection raises a `cjqr_eal_convergence_error` that retains its diagnostic audit. Capture that condition with `tryCatch()` and pass it to `cjqr_fit_diagnostics()`; this does not convert the failed fit into an accepted model.

The EAL family is more flexible than the asymmetric-Laplace family, but is still a finite-dimensional distributional model. Neither EAL flexibility nor joint estimation guarantees correct specification, consistency under every error distribution, or superior accuracy to separate quantile regressions.

Pointwise intervals for complete marginal quantile functions are available through `cjqr_eal_ci()`. That procedure bootstraps an auxiliary least-squares/residual-quantile fit, then takes the smallest interval containing both the auxiliary interval and the unchanged joint point estimate. It does not refit the joint EAL model. It is an auxiliary-bootstrap confidence enclosure, not a native EAL standard error or an EAL-bootstrap percentile interval. Its justification requires the auxiliary model's conditions, including an adequate location basis and a covariate-independent residual distribution. Coverage results from particular simulations do not establish general validity for heteroscedastic, clustered, dependent, or survey-weighted observations. The intervals are pointwise, not simultaneous bands and not prediction intervals for future outcomes.

Coefficient standard errors and a joint coefficient covariance matrix are not provided by the auxiliary-bootstrap procedure. Equation-wise summaries and smooth partial-effect plotting are described separately in `cjqr_eal_methods`. Smooth plots use `se = TRUE` by default for pointwise auxiliary-bootstrap intervals computed from centered auxiliary smooth contributions. Use `se = FALSE` for estimates alone. Neither these plots nor complete-quantile confidence requests refit the joint model. Separate experimental auxiliary-bootstrap procedures provide parametric coefficient intervals and interval-inversion p-values (`cjqr_eal_coef_ci`), whole-smooth tests (`cjqr_eal_smooth_test`), and Kendall's-tau intervals (`cjqr_eal_tau_ci`). These have their own assumptions and limitations and do not provide native joint standard errors.

`coef` extracts the two anchor-location coefficient vectors. It returns a two-column matrix when the right-hand sides, formula environments and represented designs are identical; otherwise it returns a list named by the two response variables. The stored `X` and `fit$beta` fields follow the same

matrix-versus-list convention. An equal number of columns alone does not make bases identical. `logLik` returns the unpenalized fitted log likelihood and retains `df = NA`: it does not silently attach a generally valid complexity correction. The separate `cjqr_eal_ic` interface and EAL `AIC`/`BIC` methods provide explicitly qualified conditional OPG approximations, with additional numerical checks and warnings. They do not refit the model, correct for anchor or smoothing selection, or establish comparability with native `qgam` criteria. `vcov` continues to stop rather than substitute an unsupported joint sampling covariance.

Value

An object of class `"cjqr_eal"`. It stores the selected EAL fit, marginal additive designs, constant scales and shapes, copula parameter, candidate-anchor diagnostics, controls, retained analysis information, and any explicitly supplied default `probs`. Prediction distinguishes the location predictor from the complete marginal quantile. Without intervals, a numeric-vector probability request returns quantile predictions and fitted quantiles as arrays ordered as observation by margin by probability. A list request returns a response-named list of two matrices, each ordered as observation by that margin's requested probabilities, even when the margins request the same levels or a single level. Location and marginal CDF predictions are observation-by-two matrices. For identical right-hand sides, formula environments and represented bases, `lpmatrix` returns the common observation-by-coefficient matrix; otherwise it returns two response-named matrices with their respective column counts. Confidence predictions and `confint` return the separate `cjqr_eal_ci` object. See `cjqr_eal_methods` for summary and plotting return values.

See Also

`cjqr_eal_ci`, `cjqr_eal_methods`, `cjqr_eal_ic`

Examples

```
## Not run:
dat <- cjqr_example_data(n = 200, seed = 1)
fit <- cjqr_eal(
  list(y1 ~ z + s(x, k = 6), y2 ~ z + s(x, k = 6)),
  data = dat, copula = "gaussian", anchors = 0.5, seed = 1
)
summary(fit)
cjqr_fit_diagnostics(fit)
grid <- data.frame(x = seq(-1, 1, length.out = 101), z = 0)
predict(fit, newdata = grid, type = "quantile",
        probs = c(0.1, 0.25, 0.5, 0.75, 0.9))
predict(fit, newdata = grid, type = "location")

# Different targets without refitting the joint distribution.
q_different <- predict(fit, probs = list(y1 = c(0.1, 0.5), y2 = 0.9))
q_different$y1  # observation-by-two matrix
q_different$y2  # observation-by-one matrix
ci_different <- confint(fit, probs = list(y1 = 0.1, y2 = 0.9),
                        B = 500, seed = 2)
as.data.frame(ci_different)

# Different predictor sets and basis dimensions in the two margins.
fit_distinct <- cjqr_eal(
  list(y1 ~ z + s(x, k = 6), y2 ~ s(x, k = 8)),
  data = dat, copula = "gaussian", anchors = 0.5, seed = 1
)
```

```
coef(fit_distinct)  # response-named list
predict(fit_distinct, newdata = grid, type = "lpmatrix")
plot(fit_distinct, eq = 2, select = 1)

## End(Not run)
```

`cjqr_eal_ci` *Auxiliary-bootstrap pointwise intervals for joint EAL quantiles*

Description

Constructs generally asymmetric pointwise confidence intervals that retain the original joint EAL quantile estimates. An ordinary subject-pairs bootstrap of an auxiliary unpenalized location regression provides the auxiliary interval. No joint EAL or separate quantile model is refitted.

Usage

```
cjqr_eal_ci(object, newdata = NULL,
  probs = NULL, level = 0.95,
  B = 500L, seed = 1L, chunk_size = 128L,
  max_memory_mb = 256, keep_draws = FALSE)

## S3 method for class 'cjqr_eal_ci'
print(x, ...)
```

Arguments

`object`	A strictly accepted `cjqr_eal` fit.
`newdata`	Prediction data using the fitted basis and factor levels. If `NULL`, use the original analysis rows.
`probs`	A nonempty numeric vector of distinct target probabilities strictly between zero and one, shared by both margins, or a list of two such vectors for margin-specific targets. A named list must use exactly the fitted response names and is reordered to fitted response order; an unnamed list follows that order directly. Each margin's probability order is retained. For example, `list(y1 = c(0.1, 0.5), y2 = 0.9)` requests two levels for `y1` and one for `y2`. The default `NULL` uses targets explicitly stored by `cjqr_eal`, or `c(0.1, 0.25, 0.5, 0.75, 0.9)` when none are stored, including for older fitted objects.
`level`	Pointwise confidence level, strictly between zero and one.
`B`	Number of paired bootstrap draws, an integer at least two. The default is 500. Values below 100 warn and are intended only for code checks, not inferential validation.
`seed`	Integer seed between zero and `.Machine$integer.max`. The caller's RNG kind and random state are restored.
`chunk_size`	Maximum number of prediction rows processed together when calculating endpoints. This does not change the statistical method.
`max_memory_mb`	Positive working-memory budget in MiB for a conservative preflight estimate. This is not a hard process-memory limit.

keep_draws	Retain paired row indices, auxiliary coefficient draws, and residual quantile draws? Defaults to FALSE.
x	An object returned by cjqr_eal_ci.
...	Unused arguments to the print method.

Details

Let X_j be the fixed design matrix stored for margin j by the EAL fit. The two designs may have different variables and column counts. For each margin, an unpenalized ordinary least-squares fit on its own design provides coefficients and raw residuals. The auxiliary quantile estimate is the location prediction plus the empirical residual quantile. The observed subject rows are resampled with replacement, using identical indices for both marginal designs and both outcomes. Each draw refits the two ordinary regressions and recomputes the residual quantiles. Equal-tail percentile bounds use `stats::quantile(type = 8)` for both residual quantiles and endpoints. Neither the joint model, its anchor, nor its selected smoothing parameters is refitted. The observed basis is fixed throughout. Margin-specific probability lists use one paired bootstrap over the union of the requested levels. They do not launch separate bootstrap runs by margin or level; results are returned in each margin's requested order.

If $[L_R, U_R]$ is the auxiliary interval and $\widehat{q}_J$ is the unchanged joint prediction, the reported interval is

$$[\min(L_R, \widehat{q}_J),\ \max(U_R, \widehat{q}_J)].$$

This is the smallest interval containing the auxiliary interval and the joint point estimate. It equals the auxiliary interval when the joint estimate is already inside it. Its coverage is at least the auxiliary coverage whenever the auxiliary interval is available and valid; this geometric containment does not establish the statistical validity of the auxiliary interval. The two estimators need not be independent.

The auxiliary bootstrap requires independent subject pairs and a common-error additive location model in each margin, with errors independent of the covariates and finite variance. Each finite marginal design must include one intercept, be full rank, have controlled leverage, and adequately represent the location functions. Target quantiles should be interior with positive continuous residual density. The method does not resolve finite-basis approximation bias, heteroskedasticity, covariate-dependent error shape, clustering, time dependence or informative weights. Offsets and weighted fits are outside the current interface. Adding a spatial location smooth does not account for residual spatial dependence. Strict convergence of the original fit is necessary for this interface, but does not verify these distributional assumptions.

All requested draws are attempted. If any bootstrap design is rank deficient in either margin or any draw fails numerically, a warning is issued and all interval bounds are NA. Failed draws are neither discarded nor replaced. Endpoint failures are also reported explicitly. Original joint point predictions remain unchanged even when intervals are unavailable.

These are auxiliary-bootstrap pointwise intervals, not native joint-model standard errors, simultaneous bands, or intervals for regression coefficients or copula parameters. They must not be converted to a joint covariance matrix or used as Wald standard errors. Existing simulation evidence concerns 95 percent intervals with 500 draws at probabilities 0.10, 0.25, 0.50, 0.75 and 0.90 in the studied common-error location settings; other choices are supported computationally but are not thereby empirically validated. No universal or finite-sample coverage guarantee is asserted.

Value

An object of class cjqr_eal_ci, containing:

fit, lower, upper For a numeric-vector probability request, arrays of dimension `c(n, 2, length(probs))`, where `n` is the number of prediction rows. For a list request, response-named lists of observation-by-probability matrices, with potentially different column counts for the two margins. Single columns remain matrices. `fit` is exactly the original joint quantile prediction, without recentering.

reference Auxiliary point prediction and percentile bounds with the same array or response-named-list structure as `fit`. The field name is retained for compatibility.

probs, level, method, available, error Requested probabilities, confidence level, method label, availability and any failure message. `probs` is a numeric vector for a shared request or a response-named list for margin-specific targets.

bootstrap Attempted, successful and failed draw counts, a per-draw status table, seed, fixed-basis and paired-resampling indicators, and computational diagnostics.

assumptions, interpretation Explicit applicability limitations and interpretation metadata.

draws Only if `keep_draws = TRUE`: row indices, auxiliary coefficients and residual quantiles for every attempted draw. No EAL coefficient draws are produced. For a common matrix-valued design, coefficient draws retain dimensions coefficient by margin by resample. For distinct designs, coefficient draws are a response-named list of coefficient-by-resample matrices. For shared probability requests, residual-quantile draws have probability-by-margin-by-resample dimensions. For list requests, `draws$residual_quantiles` is a response-named list of probability-by-resample matrices, with each margin's requested levels. The coefficient-draw structure depends on the marginal designs, not on whether probabilities are shared.

See Also

`predict`, `quantile`, `cjqr_eal_coef_ci` for marginal parametric coefficient intervals, including intercepts.

Examples

```
## Not run:
dat <- cjqr_example_data(n = 200, seed = 1)
fit <- cjqr_eal(list(y1 ~ z + s(x, k = 6), y2 ~ z + s(x, k = 6)),
                data = dat, anchors = 0.5, seed = 1)
grid <- data.frame(x = seq(-1, 1, length.out = 51), z = 0)
ci <- cjqr_eal_ci(fit, newdata = grid, probs = c(0.1, 0.5, 0.9),
                  B = 500, seed = 2)
print(ci)
head(as.data.frame(ci))

## End(Not run)
```

`cjqr_eal_coef_ci` *Auxiliary-bootstrap inference for marginal parametric effects*

Description

Experimental, opt-in coefficient inference: finite-sample calibration has not been established and validation found undercoverage in some settings. Adds confidence intervals and approximate interval-inversion tests to the unchanged marginal parametric effects of a certified joint EAL fit. Only auxiliary ordinary regressions are bootstrapped; the joint likelihood is never refitted. No joint-model standard errors are produced.

Usage

```
cjqr_eal_coef_ci(object, level = 0.95, B = 500L, seed = 1L,
  max_memory_mb = 256, keep_draws = FALSE)
## S3 method for class 'cjqr_eal_coef_ci'
print(x, digits = 4L,
  signif.stars = getOption("show.signif.stars", TRUE), ...)
```

Arguments

object	A certified cjqr_eal model.
level	Confidence level strictly between zero and one.
B	Number of paired-subject bootstrap draws; at least two. Values below 100 are only for code checks.
seed	Nonnegative integer seed. The caller's RNG state is preserved.
max_memory_mb	Positive estimated working-memory limit in MiB.
keep_draws	Retain paired row indices, auxiliary coefficient draws and anchor-specific auxiliary intercept draws?
x	An object returned by cjqr_eal_coef_ci.
digits	Number of printed significant digits.
signif.stars	Show conventional significance symbols and their legend for available experimental auxiliary-bootstrap tests? Unavailable tests remain NA without stars.
...	Unused printing arguments.

Details

Under a common-error additive location model with finite-variance errors independent of covariates, conditional means and quantiles share their non-intercept covariate effects. For each margin an unpenalized ordinary regression uses the original fitted design, including all smooth columns. Independent subjects are resampled with replacement, using identical row indices for both outcomes. Each auxiliary regression is refitted on that sample. This is not a bootstrap of the joint EAL estimator.

Let $[L_R, U_R]$ be the equal-tail percentile interval for the auxiliary coefficient, using `quantile(type = 8)`, and let b_J be the unchanged joint estimate. The reported interval is

$$[\min(L_R, b_J), \max(U_R, b_J)].$$

It contains the auxiliary interval, so inherits at least its coverage when that interval is valid for the same population coefficient. Neither independence of the two estimators nor a covariance estimate for the joint estimator is needed.

Tests concern the null coefficient zero. The p-value is obtained by inverting the nested percentile-hull intervals, with strict exclusion of zero. If the joint estimate is positive, inversion checks the lower auxiliary endpoint; if negative it checks the upper endpoint. A zero joint estimate gives p-value one. For numerical resampling resolution, reported p-values are floored at $2/(B + 1)$. This floor is conservative relative to the raw inversion; it is not an exact Monte Carlo test or a finite-sample validity correction. At very high confidence levels the p-value floor can be more conservative than the displayed interval; increase B if a warning reports insufficient resolution.

These are not Wald tests. A confidence-interval width must not be converted into a joint standard error. The tests cannot reject unless the auxiliary percentile test rejects and cannot demonstrate

more powerful inference from joint fitting. Tests across coefficients or margins are not multiplicity-adjusted.

Marginal intercepts target the fitted anchor quantile, not the mean. Their auxiliary estimate is the OLS intercept plus the residual quantile at that anchor, recomputed using the resampled observations on each bootstrap draw. All draws use the original basis and its centering constraints. The original joint intercept is retained and the interval is the hull defined above.

If the prespecified candidate grid contains K anchors, auxiliary intercept intervals use tail probabilities $\alpha/(2K)$ and $1 - \alpha/(2K)$, where $\alpha = 1 - \text{level}$. Intercept p-values are $\min(1, Kp_0)$, with p_0 the unadjusted interval-inversion p-value, and resolution floor $\min(1, 2K/(B+1))$. This Bonferroni protection includes failed and unselected candidate anchors; it is not conditional selective inference. The selected-target coverage guarantee is inherited from simultaneous auxiliary coverage across candidate anchors within each margin, not across all coefficients or both margins. A single prespecified anchor uses $K = 1$. Missing or inconsistent anchor metadata causes an error, rather than silently treating the selected anchor as fixed.

The anchor grid must have been specified before observing outcomes, and the residual distribution must have positive, finite, continuous density near every candidate quantile. Selection protection does not repair a poorly calibrated auxiliary bootstrap. In particular, a displayed marginal intercept refers to the selected anchor and original spline-centering constraints; it is not automatically a median or mean intercept.

The fixed basis must adequately represent the location functions, have full column rank and controlled leverage. The procedure does not resolve spline approximation bias, clustering, residual spatial dependence, covariate-dependent error distributions or informative weights. Percentile-bootstrap validity is asymptotic, not guaranteed for every finite sample. All bootstrap draws must succeed; otherwise all intervals and tests are unavailable, with failed draws neither discarded nor replaced.

Only marginal parametric columns are tested. Smooth-basis coefficients and all dependence parameters, including the Equation 3 intercept, are excluded. The two margins may have different covariates and basis dimensions.

Value

An object of class `cjqr_eal_coef_ci`, with `tables` containing joint estimates, confidence limits and auxiliary-bootstrap p-values for each margin; `reference` containing the auxiliary estimates, percentile limits and auxiliary interval-inversion p-values (the `reference` field name is retained for compatibility); `bootstrap` containing draw status, timing and resolution metadata; `intercept` describing the full anchor grid, selected anchor and within-margin selection protection; `available`, `error`, `assumptions` and `interpretation`. With `keep_draws = TRUE`, `draws` contains row indices and raw auxiliary OLS parametric coefficient draws, and an `intercepts` array (anchor by margin by draw) of OLS intercepts plus residual quantiles. These are not EAL coefficient draws.

See Also

`cjqr_eal_ci`, `cjqr_eal_methods`

Examples

```
## Not run:
dat <- cjqr_example_data(n = 200, seed = 1)
fit <- cjqr_eal(list(y1 ~ z + s(x, k = 6), y2 ~ z + s(x, k = 6)),
                data = dat, anchors = 0.5, seed = 1)
summary(fit, inference = "auxiliary", B = 500, seed = 1)
ci <- cjqr_eal_coef_ci(fit, B = 500, keep_draws = TRUE)
```

```
ci$tables

## End(Not run)
```

`cjqr_eal_control` *Validate numerical overrides for the EAL fitting interface*

Description

Constructs a named list of EAL numerical control overrides. Unknown names, inappropriate copula bounds, invalid initial dependence parameters and unsafe convergence tolerances are rejected before fitting.

Usage

```
cjqr_eal_control(..., copula = "gaussian")
```

Arguments

`...`	Uniquely named overrides accepted by `cjqr_eal()`: iteration limits, numerical tolerances, dependence initialization/search bounds, `shape_bound`, and `trace`. See `cjqr_eal` for the full list. No unnamed arguments are accepted.
`copula`	The copula family whose bound controls are being validated. Use the same family when fitting. This validates the controls; it does not set or estimate the fitted copula.

Details

This constructor does not run a model or alter random state. It does not change the fixed degree, anchor-selection rule, Gaussian-REML penalty source, or strict numerical certification requirements. The fixed Student-t degrees of freedom must be passed as the separate `df` argument of `cjqr_eal()`, not inside this control list.

Value

An ordinary named list containing only the supplied overrides. With no overrides, returns `list()`, retaining all model defaults.

See Also

`cjqr_eal`, `cjqr_fit_diagnostics`

Examples

```
cjqr_eal_control(maxit = 1500L)
cjqr_eal_control(frank_bounds = c(-20, 20), copula = "frank")
```

`cjqr_eal_ic` *Conditional OPG information-criterion approximations for joint EAL fits*

Description

Computes a conditional OPG/Fisher AIC-type approximation and an effective-degrees-of-freedom BIC-type heuristic from an existing joint EAL fit. The calculation uses the normalized, unpenalized joint density and all fitted parameter directions, without joint refitting. These are qualified model-comparison diagnostics, not unconditional information criteria or validated evidence measures.

Usage

```
cjqr_eal_ic(object, warn = TRUE)

## S3 method for class 'cjqr_eal_ic'
print(x,
  digits = max(3L, getOption("digits") - 2L), ...)
## S3 method for class 'cjqr_eal'
AIC(object, ..., k = 2, warn = TRUE)
## S3 method for class 'cjqr_eal'
BIC(object, ..., warn = TRUE)
```

Arguments

`object`	A fitted `cjqr_eal` object with complete, consistent convergence evidence and an admissible conditional score information matrix.
`warn`	Logical. The default `TRUE` warns about the conditional approximation and its limitations. `FALSE` suppresses that warning; it changes neither the calculation nor the assumptions.
`x`	An object returned by `cjqr_eal_ic`.
`digits`	Number of significant digits used when printing.
`k`	A positive finite multiplier of conditional effective degrees of freedom. The default `2` gives the AIC-type criterion.
`...`	For `AIC` and `BIC`, additional `cjqr_eal` objects satisfying the comparison restrictions below. For printing, additional generic-method arguments, currently unused.

Details

Let s_i be the full unpenalized joint-density score for independent subject pair i, $J = \sum_i s_i s_i^\mathsf{T}$, and P the fitted penalty Hessian on the same parameterization and summed-likelihood scale. The reported conditional complexity and criteria are

$$d_{\mathrm{OPG}} = \mathrm{tr}\{(J+P)^{-1}J\},$$

$$\mathrm{AIC}_{\mathrm{OPG}} = -2\ell + 2d_{\mathrm{OPG}}, \qquad \mathrm{BIC}_{\mathrm{EDF}} = -2\ell + \log(n)d_{\mathrm{OPG}}.$$

Here ℓ is the fitted unpenalized joint log likelihood and n is the number of retained subject pairs, not twice that number. The `AIC` method substitutes `k` for 2 when requested. The full score includes both marginal regression blocks and 13 additional directions: two log scales, ten marginal shape

parameters, and one copula dependence parameter. The fixed Student-t copula degrees of freedom are not an estimated parameter and do not add a score direction.

The selected common anchor, fixed EAL degree, selected smoothing parameters, and supplied copula degrees of freedom are conditioned upon. Their selection uncertainty is not included in the effective degrees of freedom. In particular, this is not the smoothing-parameter-uncertainty correction of Wood, Pya and Saefken (2016). GJRM uses likelihood-Hessian/Fisher-based information in its penalized-likelihood calculations; this implementation uses the outer product of subject scores as an expected-Fisher approximation. It does not reproduce GJRM's information matrix or claim identical criteria.

The interpretation requires independent subject pairs, a correctly specified working joint density, and an identified interior solution away from fitted density cusps. Numerical checks alone do not establish these statistical assumptions. The EDF BIC expression is a heuristic: it has not been established here as a Laplace marginal-likelihood approximation, a consistent selection criterion, or a Bayes factor. Neither criterion corrects for model misspecification, anchor selection, smoothing selection, or a search over fixed copula degrees of freedom. If different fixed degrees of freedom are compared, each result remains conditional on its supplied value.

Multiple-object comparisons are deliberately scoped to copula comparisons. They require identical marginal data, paired responses and row order, retained subject rows, marginal design matrices, EAL degree, candidate-anchor sets, and fitted penalty matrices. The selected anchor may differ, but the calculation conditions on each selection without charging for that search. A changed response transformation or sample does not give a comparable likelihood. These methods do not support mixed comparisons with `qgam` or other model classes. Native `qgam` AIC/BIC values arise from a calibrated working ELF loss/density and must not be ranked against these joint-density criteria. They also do not select the most appropriate quantile probability.

The method independently replays the joint log densities, penalty and penalized objective against the stored fit. It stops for incomplete or inconsistent convergence evidence, invalid density information, fitted cusps, search boundaries, singular or unsafe penalized information, inadmissible trace values, or failed replay. There is no parameter-count, ridge-regularized or initializer-based fallback. No fit is changed or refitted. `logLik(fit)` continues to return `df = NA`; these qualified complexity estimates are available through this explicit route.

Value

`cjqr_eal_ic` returns a list of class `"cjqr_eal_ic"` with the unpenalized `logLik`, conditional `effective_df`, represented `parameter_dimension`, paired `nobs`, `AIC`, `BIC`, copula specification, selected anchor, degree, conditioning assumptions, limitations, numerical diagnostics, and `joint_refits = 0L`. Its `statistical_validity_established` field is `FALSE`; successful numerical checks do not validate model-selection performance. Printing displays the criteria and their qualifications and returns the object invisibly. For one fit, `AIC` and `BIC` return a numeric value and, by default, the qualification warning. For several compatible fits, they return a data frame with conditional `df` and the requested criterion, plus method and qualification attributes.

References

Wood, S. N., Pya, N. and Saefken, B. (2016). Smoothing parameter and model selection for general smooth models. *Journal of the American Statistical Association*, 111, 1548–1575. doi:10.1080/01621459.2016.1180986.

Fasiolo, M., Wood, S. N., Zaffran, M., Nedellec, R. and Goude, Y. (2021). qgam: Bayesian Nonparametric Quantile Regression Modeling in R. *Journal of Statistical Software*, 100(9), 1–31. doi:10.18637/jss.v100.i09.

GJRM package source, postVb, Hessian-based influence matrix and total EDF calculation: https://raw.githubusercontent.com/cran/GJRM/master/R/postVb.r.

See Also

cjqr_eal, cjqr_fit_diagnostics, AIC, logLik.gam

Examples

```
## Not run:
dat <- cjqr_example_data(n = 200, seed = 1)
formulas <- list(y1 ~ z + s(x, k = 6), y2 ~ z + s(x, k = 6))
fit <- cjqr_eal(formulas, data = dat, copula = "gaussian",
                anchors = 0.5, seed = 1)
# Criteria require additional information-matrix checks and may be unavailable.
ic <- cjqr_eal_ic(fit)
ic$effective_df
ic$diagnostics
AIC(fit)  # conditional OPG AIC-type approximation, with a warning
BIC(fit)  # EDF BIC-type heuristic, with a warning

# Use the same paired data, designs, anchor candidates and penalty policy.
fit_other <- cjqr_eal(formulas, data = dat, copula = "student_t", df = 5,
                      anchors = 0.5, seed = 1)
AIC(fit, fit_other)
BIC(fit, fit_other)

## End(Not run)
```

---

cjqr_eal_methods *Equation-oriented summaries and plots for joint EAL models*

Description

Presents the joint EAL estimates equation by equation and plots centered smooth contributions with auxiliary-bootstrap pointwise intervals by default. Equation and smooth selection conventions follow the presentation used by GJRM, without borrowing uncertainty estimates from the Gaussian models used to prepare the basis and penalties. Complete marginal quantile profiles remain available as a separate plot type, optionally with auxiliary-bootstrap pointwise intervals.

Usage

```
## S3 method for class 'cjqr_eal'
summary(object, inference = c("none", "auxiliary", "reference"),
  level = 0.95, B = 500L, seed = 1L, max_memory_mb = 256, ...)
## S3 method for class 'summary.cjqr_eal'
print(x,
  digits = max(3L, getOption("digits") - 3L),
  signif.stars = getOption("show.signif.stars", TRUE), ...)
## S3 method for class 'cjqr_eal'
print(x, ...)
## S3 method for class 'cjqr_eal'
plot(x, eq = 1L, select = NULL,
```

```
type = c("smooth", "quantile"), newdata = NULL,
variable = NULL, probs = NULL, margin = NULL,
interval = c("none", "confidence"), level = 0.95,
B = 500L, seed = 1L, se = TRUE, n = 100L, n2 = 40L,
pages = 0L, rug = TRUE, scheme = 0L, too.far = 0.1,
draw = TRUE, max_memory_mb = 256, ...)
```

Arguments

object, x	A certified cjqr_eal fit; for the summary print method, the result of summary.cjqr_eal.
digits	Number of significant digits for printed numeric estimates.
signif.stars	Display R/GJRM-style significance symbols and a legend for available auxiliary-bootstrap p-values. Uses the R option show.signif.stars, or TRUE when that option is unset. These symbols do not establish coefficient calibration or native joint SEs.
inference	For summaries, "none" retains estimate-only tables without bootstrap computation. "auxiliary" adds auxiliary-bootstrap intervals and p-values for marginal parametric effects using cjqr_eal_coef_ci. The legacy "reference" option remains an alias for "auxiliary".
max_memory_mb	Estimated memory limit, in megabytes, for auxiliary-bootstrap inference.
eq	Marginal equation to plot, either 1 or 2. The constant dependence equation has no smooth component to plot.
select	One or more distinct smooth indices within the selected equation, in the row order of the printed smooth table. The corresponding Select values remain available in summary(fit)$tableNP1 and summary(fit)$tableNP2. NULL selects all its smooths. Used only with type = "smooth".
type	"smooth" plots centered partial smooth contributions; "quantile" plots complete marginal quantile predictions.
newdata	Prediction data for complete quantile plots, or NULL for the original analysis data. Not used for smooth plots.
variable	Numeric prediction covariate used for the horizontal axis of a complete quantile plot. If NULL, use analysis-row order.
probs	A nonempty numeric vector of distinct probabilities strictly between zero and one, shared by both margins, or a list of two such vectors for margin-specific complete quantile predictions. Named lists must use exactly the fitted response names; unnamed lists follow fitted response order. The selected equation's probabilities determine its curves and legend, in supplied order. NULL uses targets explicitly stored at fitting, or the historical plotting default c(0.1, 0.5, 0.9) when none are stored, including for older objects. Unused for smooth plots.
margin	Compatibility alias for eq; conflicting selections are rejected. Supplying it without type selects quantile plotting.
interval	For complete quantiles, "confidence" calls cjqr_eal_ci; their default is "none". For smooth plots, an explicitly supplied value is an alias for se: "confidence" requests pointwise intervals and "none" requests estimates alone. If both arguments are supplied, they must agree.
level, B, seed	Confidence level, auxiliary bootstrap draw count and seed for the auxiliary bootstrap. Defaults are 95% intervals, 500 draws and seed 1; no joint model is refitted. The caller's random-number state is restored after the bootstrap.

se	For smooth plots, include auxiliary-bootstrap pointwise confidence intervals? Defaults to TRUE; use FALSE for estimates alone. Despite the conventional argument name, these intervals are not based on joint smooth standard errors or Gaussian Wald approximations. Complete-quantile plots use interval instead.
n	Grid size for a one-dimensional smooth, from 2 to 10000.
n2	Grid size along each axis of a two-dimensional smooth, from 2 to 250.
pages	Number of pages for smooth panels. Zero leaves the current graphics layout unchanged; positive values arrange panels and restore the previous layout afterwards. A two-dimensional smooth with confidence intervals contributes three panels: estimate, lower bound and upper bound.
rug	Add observed covariate locations to smooth plots?
scheme	Two-dimensional smooth display: 0 for contours, 1 for a perspective surface, or 2 for a heatmap with contours.
too.far	Nonnegative distance used by mgcv::exclude.too.far to mask unsupported two-dimensional grid locations. Zero disables masking.
draw	If FALSE, return plot data or predictions without opening or drawing to a graphics device.
...	For plotting, ordinary graphical arguments such as xlab, ylab, main, col, lwd and axis limits. For complete quantile plots, col, lwd and lty must each have length one or the number of probabilities for the selected margin and are applied to the actual curves and their legend entries in probability order. A scalar is repeated; other lengths are rejected. For printing a fit, arguments passed to its summary print method. Other summary arguments are not supported. Gaussian GAM inference, partial residuals and transformations of the displayed term are not supported through these arguments.

Details

One-dimensional smooth plots use the smooth term label, for example s(age), as the default y-axis label. Two-dimensional smooths retain their covariate axis labels; perspective surfaces use the smooth label on the z-axis. User-supplied axis labels override these defaults. These labels do not change the centring of the smooth contributions.

The summary separates parametric coefficient estimates from smooth components in each marginal equation. Printed headings are simply EQUATION 1, EQUATION 2 and EQUATION 3; responses remain in their formulas. Smooths are printed with columns edf and p-value, with significance symbols. EDF is a descriptive conditional joint OPG trace, not a native sampling covariance or test reference df. P-values are available with inference = "auxiliary"; test dimensions and statistics are retained in smooth_inference but are not printed; see cjqr_eal_smooth_test. Smooth labels form the row names. Selection indices and represented coefficient dimensions remain available in tableNP1 and tableNP2 for compatibility. The equations may have different covariates, smooth terms and basis sizes; the summary and plot indices refer to the selected equation's own basis. For example, select = 2 in equation 2 need not identify a term that exists in equation 1. Coefficient dimension is not an effective degrees of freedom or reference degrees of freedom estimate and is never inserted in those columns. Equation 3 reports the intercept on the dependence-link scale. The link is labelled for the natural copula parameter θ: atanh for Gaussian and Student-t, log for Clayton, identity for Frank, and log(theta - 1) for Gumbel. The compact footer reports Kendall's tau transformed from the fitted copula parameter, without printing the natural-scale θ value. For Gaussian and Student-t copulas, θ is the correlation parameter; it is not the displayed link-scale intercept. A Student-t copula's fixed degrees of freedom are displayed but are not estimated.

The methods do not provide joint coefficient covariance or Wald tests. Conditional EDF is withheld at likelihood cusps, boundaries or unsafe information matrices. Significance stars only summarize the existing experimental auxiliary-bootstrap p-values; unavailable tests receive none. In particular, the Gaussian preparation models' covariance matrices and effective degrees of freedom do not describe the joint EAL estimator and are not displayed or used to generate bands.

Coefficient inference is explicitly opt-in: use `summary(fit, inference = "auxiliary")`. Its marginal parametric intervals and p-values inherit protection from the auxiliary location estimator under its assumptions; they are not native joint SEs or Wald tests. Auxiliary marginal intercept estimates combine an OLS intercept and residual quantile, with Bonferroni protection across the complete prespecified anchor grid within each margin. Whole-smooth tests share the same auxiliary draws with HC3-studentized calibration. An auxiliary-bootstrap Kendall's tau interval uses the same paired residual bootstrap; see `cjqr_eal_tau_ci`. Native theta-scale inference and joint covariance remain unavailable. See `cjqr_eal_coef_ci` for the parametric construction, limitations and conservative bootstrap p-value resolution floor.

The footer reports only `Convergence: converged`, `Convergence: not converged`, or `Convergence: not assessed`. The status is based on stored numerical certificates and the anchor-selection audit. Missing details are not assessed, and contradictions or failed checks are not converged, rather than unconditionally claiming convergence. These checks do not prove a global optimum or statistical validity. Scaled stationarity, the threshold, individual checks, reasons and boundaries are retained in the returned `convergence` component; its `status` uses the same three labels. Verbose inference notes remain in the returned object and documentation but are not printed in the footer.

For a smooth with coefficient block $\widehat{\beta}_s$, plotted values are its fixed-basis contribution minus its empirical mean over the original paired analysis rows. Thus all centering uses the original observed covariates, not the plotting grid. The values come from the joint EAL coefficients, not the Gaussian preparation coefficients. In the common-error location model these centered covariate effects do not depend on the target quantile; full quantiles differ through their intercept shifts.

With `se = TRUE`, complete paired rows are resampled and auxiliary unpenalized least-squares regressions are fitted using the original full marginal design matrices. The selected smooth coefficients are projected onto its evaluation basis, with centering over the original cohort in every draw. Equal-tail percentile endpoints (quantile type 8) are then enlarged, if necessary, to contain the unchanged joint smooth estimate. Only smooth contributions enter these intervals: no intercept or residual quantile is added. They do not use complete-quantile intervals, Gaussian preparation-model covariance or refits of the joint EAL model.

The intervals require independent subject pairs, finite-variance common location errors independent of covariates, and an adequate full-rank fixed basis. Containment inherits coverage from a valid auxiliary interval; it does not remove basis bias or guarantee nominal finite-sample coverage. Intervals are pointwise, not simultaneous bands for the entire smooth. All requested draws must be valid. A rank-deficient or otherwise invalid draw produces an error rather than being discarded or replaced. Smooth confidence plots do not support non-unit observation weights; estimates can still be plotted with `se = FALSE`.

One-dimensional plots display two dashed confidence-limit lines, without shading. The bounds use the smooth curve's colour and half its line width; the estimated curve is solid by default. For two-dimensional smooths, the estimate, lower confidence bound and upper confidence bound are shown as separate surfaces using the same surface-value scale. Unsupported grid locations are masked in all three surfaces. Use `pages = 1` to arrange them on one page.

Automatic smooth plots support one or two nonconstant numeric covariates, including ordinary tensor-product smooths. Numeric by-variable curves are evaluated with that variable set to one; factor-by curves use the smooth's factor level. Such profiles are identified in returned data and plot labels. Unsupported factor-only, random-effect, MRF or higher-dimensional smooths are rejected rather than converted to misleading curves. A two-dimensional coordinate smooth can occur in only one margin. Plotting such a location effect does not imply a residual spatial-dependence model.

For complete quantile plots, predictions are connected in horizontal-axis order. To obtain a conditional profile, supply `newdata` that varies the chosen covariate and holds the other covariates fixed; the function does not silently fix them. Prediction data must include the union of covariates used in both marginal formulas, even when only one margin is displayed. A plot against a covariate in the original data can therefore also reflect variation in other covariates. Optional auxiliary-bootstrap intervals cover complete quantiles pointwise under their documented assumptions. They are not confidence bands for the isolated smooth and are never repurposed as such. For example, `probs = list(y1 = c(0.1, 0.5), y2 = 0.9)` with `eq = 2` draws only the 0.9 quantile of response `y2`, with a matching legend. A numeric vector still requests the same levels for both margins. Changing these targets does not refit the EAL distribution or alter its common selected anchor.

The default quantile curves retain sequential palette colors, line width two and solid lines. Custom widths must be positive and finite; line types may be R codes 0 through 6, standard names or even-length hexadecimal patterns. Confidence bounds share the corresponding curve color and use half its line width. They use a dashed pattern, or a long-dashed pattern when the curve itself uses the standard dashed pattern, so the bounds remain distinct from the quantile estimate. Styling changes no prediction or interval values.

For compatibility with earlier complete-quantile plotting calls, omitting `type` while supplying `newdata`, `variable`, `probs` or `margin` selects quantile plotting. An explicit `interval` also selects quantile plotting when neither `eq` nor `select` is supplied. A bare `plot(fit)` defaults to smooths in equation 1. Use an explicit `type` when there is any ambiguity.

Value

The summary is a `summary.cjqr_eal` list containing equation metadata, marginal parametric tables `tableP1` and `tableP2`, an estimate-only dependence table `tableP3`, smooth descriptions `tableNP1` and `tableNP2`, and display matrices in `smooth_tables`. The marginal parametric tables contain estimates by default; `inference = "auxiliary"` adds available auxiliary-bootstrap intervals and p-values. The result also contains the fitted dependence parameter, Kendall's tau, numerical diagnostics and, when requested, `coefficient_inference`, `smooth_inference` and `tau_ci`. Unavailable inference is explicitly recorded. Print methods return their argument invisibly.

Smooth plotting invisibly returns a named list of plot data, including grids, centered joint contributions, centering constants, centered training contributions, selected coefficient indices and excluded grid locations. The `se` field is `NULL`. Quantile plotting invisibly returns the corresponding prediction array, response-named list of prediction matrices for a list probability request, or `cjqr_eal_ci` object when confidence intervals are requested. With smooth confidence intervals, each panel also contains `lower`, `upper`, `reference_lower`, `reference_upper`, `level`, `interval` and `bootstrap`. The `se` field remains `NULL` because a joint-model standard error is not calculated. The `reference_lower` and `reference_upper` names are retained for compatibility and contain the auxiliary interval endpoints.

References

GJRM plotting and summary documentation: https://search.r-project.org/CRAN/refmans/GJRM/html/plot.SemiParBIV.html and https://search.r-project.org/CRAN/refmans/GJRM/html/summary.SemiParBIV.html.

See Also

`cjqr_eal`, `cjqr_eal_ci`, `plot.gam`, `PredictMat`

Examples

```
## Not run:
```

```
dat <- cjqr_example_data(n = 200, seed = 1)
fit <- cjqr_eal(
  list(y1 ~ z + s(x, k = 6), y2 ~ z + s(x, k = 6)),
  data = dat, anchors = 0.5, seed = 1
)
plot(fit, eq = 1, select = 1)  # 95
plot(fit, eq = 2, select = 1, se = FALSE)  # estimates only
smooth_data <- plot(fit, eq = 1, select = 1, draw = FALSE)
smooth_data[[1]]$lower

## End(Not run)
```

`cjqr_eal_smooth_test` *Auxiliary-bootstrap tests of complete marginal smooth effects*

Description

Experimental whole-smooth tests for certified joint EAL fits. Only auxiliary ordinary least-squares regressions are resampled; the joint model is not refitted. The reported joint smooth is unchanged. This does not provide native joint standard errors, Wald tests, or tests of nonlinearity alone.

Usage

```
cjqr_eal_smooth_test(object, level = .95, B = 500L, seed = 1L,
  max_memory_mb = 256, keep_draws = FALSE)
## S3 method for class 'cjqr_eal_smooth_test'
print(x, digits = 4L,
  signif.stars = getOption("show.signif.stars", TRUE), ...)
```

Arguments

`object`	A certified `cjqr_eal` fit.
`level`	Confidence-region level, strictly between zero and one.
`B`	Number of paired bootstrap resamples; at least two and more than the tested block dimension. Values below 100 are only for code checks.
`seed`	Nonnegative integer seed; caller RNG is preserved.
`max_memory_mb`	Estimated working-memory limit in MiB.
`keep_draws`	Retain paired indices and full auxiliary OLS coefficients?
`x`	An object returned by this function.
`digits`	Number of printed significant digits.
`signif.stars`	Print conventional p-value significance symbols?
`...`	Unused printing arguments.

Details

Under the common-error additive location model, the centered smooth coefficient block has the same target in the conditional mean and all marginal quantiles. For each margin, OLS uses the original full design, including all other smooths and parametric effects. Both margins use the same resampled subject indices; the original basis and its centering constraints are fixed.

Let r be an OLS smooth coefficient block, b the unchanged joint block, and V the block of the full OLS HC3 sandwich covariance. HC3 is an auxiliary covariance, not a joint EAL covariance. Each bootstrap draw recomputes OLS and HC3, including leverages from the resampled design, to obtain r^* and V^*. Radial calibration uses

$$Q^* = (r^* - r)^\top (V^*)^{-1}(r^* - r).$$

The auxiliary ellipsoid is enlarged by taking its convex hull with b. For nonzero b, exclusion of the zero block is assessed by

$$T = \min_{t \geq 0} (r + tb)^\top V^{-1}(r + tb).$$

For a zero joint block, the statistic is zero and p-value one. Otherwise the bootstrap p-value is $(1 + \#\{Q^* \geq T\})/(B + 1)$. The added one limits resampling resolution; it is not an exact Monte Carlo test or a guarantee of finite-sample validity. Studentization accounts for covariance variability within each resample. No native joint Wald distribution is asserted.

The hull contains the auxiliary region, so its rejection set is a subset of the auxiliary test's. Its inference cannot be more powerful than the auxiliary procedure. Multiplying a nonzero joint block by a positive scalar leaves the hull statistic unchanged; this is a geometric containment test, not inference on the magnitude of the joint estimator's sampling variation.

The null is that the entire identifiable centered smooth is zero, including its unpenalized linear component. Significance is not evidence of nonlinearity alone. Calibration is simultaneous within a coefficient block, not across smooths or margins. No anchor multiplicity correction is needed because the centered smooth target is anchor-invariant under the common-error assumptions. The model formulas and tested blocks must not have been selected by searching for significance in the same data.

The auxiliary bootstrap requires independent subjects, finite-variance common errors independent of covariates, an adequate fixed full-rank basis, controlled leverage, and nonsingular block covariance matrices. All auxiliary resamples and their HC3 calculations must succeed. Failed draws are not dropped or replaced; smooth tests are unavailable on failure. Singular or ill-conditioned covariances are not repaired by ridge or arbitrary rank truncation.

`summary(fit, inference = "auxiliary")` shares the paired resamples with parametric inference. Summary smooth columns are `edf` and `p-value`, with significance symbols. The test dimension and statistic remain available in the returned inference tables and details, but are not printed. The dedicated test function prints p-values and significance symbols without computing EDF. Internally, Rank is the tested block dimension, not GJRM/mgcv's adjusted `Ref.df`; the statistic is not a native joint `Chi.sq`.

Summary EDF is a separate descriptive conditional joint OPG trace. Full per-subject scores include marginal coefficients, scales, shapes and dependence. For score information J and the full fitted penalty P, a term's EDF is the corresponding block trace of $(J + P)^{-1}J$. The selected anchor, degree, smoothing parameters and Student-t copula degrees of freedom are held fixed. This is a model-complexity approximation, not a sampling covariance or the reference degrees of freedom for these tests. EDF is withheld at detected nondifferentiable fitted residuals, numerical search boundaries, or unsafe information matrices. No optimizer Hessian or Gaussian initializer EDF is used.

Value

An object of class `cjqr_eal_smooth_test`, containing numeric marginal `tables`, per-block `details`, `available`, `bootstrap` status and resolution metadata, and `interpretation`. Details include HC3 auxiliary covariance, studentized radial draws, conditioning diagnostics and the auxiliary-only statistic and p-value. Optional `draws` contains auxiliary OLS coefficients, not joint EAL draws.

See Also

`cjqr_eal_coef_ci`, `cjqr_eal_methods`

Examples

```
## Not run:
dat <- cjqr_example_data(n = 200, seed = 1)
fit <- cjqr_eal(list(y1 ~ z + s(x, k = 6), y2 ~ z + s(x, k = 6)),
                data = dat, anchors = 0.5, seed = 1)
summary(fit, inference = "auxiliary", B = 1999, seed = 1)
cjqr_eal_smooth_test(fit, B = 1999, seed = 1)

## End(Not run)
```

---

cjqr_eal_tau_ci *Auxiliary-bootstrap confidence interval for residual-copula Kendall's tau*

Description

An experimental interval for constant Kendall's tau under the common-error location model. The fitted copula-derived point estimate is unchanged. Only auxiliary OLS models are bootstrapped; the joint EAL model is not refitted.

Usage

```
cjqr_eal_tau_ci(object, level = .95, B = 500L, seed = 1L,
  max_memory_mb = 256, keep_draws = FALSE)
## S3 method for class 'cjqr_eal_tau_ci'
print(x, digits = 4L, ...)
```

Arguments

`object`	A certified `cjqr_eal` fit.
`level`	Confidence level strictly between zero and one.
`B`	Number of paired subject resamples, at least two. Values below 100 are only for code checks.
`seed`	Nonnegative integer seed. Caller RNG state is preserved.
`max_memory_mb`	Estimated working-memory limit in MiB.
`keep_draws`	Retain the auxiliary bootstrap tau values?
`x`	An object returned by this function.
`digits`	Number of significant digits to print.
`...`	Unused printing arguments.

Details

Each marginal auxiliary OLS fit uses the complete original fixed design, which may differ between margins. For paired residuals e_{i1}, e_{i2}, the auxiliary statistic is Kendall's tau-a,

$$\widehat{\tau}_R = \frac{1}{\binom{n}{2}} \sum_{i<k} \operatorname{sign}(e_{i1} - e_{k1}) \operatorname{sign}(e_{i2} - e_{k2}).$$

Each paired subject bootstrap refits both auxiliary OLS models and computes this statistic from its in-bag residuals. Duplicate subjects are retained and tied pair kernels contribute zero; the denominator remains $\binom{n}{2}$. The implementation uses the concordance counts in the public `survival::concordancefit` API, without using its standard errors or any independence-null variance formula.

Let $[q_L, q_U]$ be the type-8 percentile interval from these auxiliary draws. The basic bootstrap interval is $[2\widehat{\tau}_R - q_U, 2\widehat{\tau}_R - q_L]$. Use the fixed envelope

$$[L_R, U_R] = [\min(q_L, 2\widehat{\tau}_R - q_U), \max(q_U, 2\widehat{\tau}_R - q_L)] \cap [-1, 1].$$

This contains both component intervals within the tau parameter space. It guards against choosing a bootstrap bias direction or a procedure separately for each dataset. It is intentionally conservative, without a tuned critical-value multiplier; it is not an exact or higher-order validity claim. For $\widehat{\tau}_J$, the fitted copula parameter transformed to Kendall's tau, the reported interval is

$$[\min(L_R, \widehat{\tau}_J), \max(U_R, \widehat{\tau}_J)].$$

This interval contains the auxiliary interval and the unchanged joint point estimate. Its coverage protection is inherited from the approximate auxiliary interval. It is not a native joint-parameter covariance interval, is generally asymmetric, and need not be shorter than the auxiliary interval.

Assumptions include independent subjects, continuous errors independent of covariates, a covariate-constant copula, finite error variance, and an adequate full-rank fixed location basis with controlled leverage. The target is the true residual-copula Kendall's tau. This is not inference for a misspecified likelihood's pseudo-true copula parameter. The auxiliary statistic is invariant to constant marginal shifts and positive scales; no anchor selection correction is needed. Covariate or basis selection and clustered observations are not accounted for.

Original residual ties are not supported for this continuous-copula target. Bootstrap ties caused by duplicated subjects are expected and supported. All draws must succeed; no failed draw is dropped or replaced. Original perfect rank dependence and zero bootstrap variability give unavailable intervals rather than spurious zero-width confidence intervals. Intervals use only the universal tau bounds, not a fitted family's restricted sign range.

`summary(fit, inference = "auxiliary")` shares the same paired resamples with coefficient and smooth inference and reports this interval in its compact footer. A default summary does not resample and marks the CI as not computed. The standalone function does not require a stored candidate-anchor grid.

Value

An object of class `cjqr_eal_tau_ci` with `estimate`, `interval`, `level`, `available`, `reason`, the original `reference` statistic and pair counts, `reference_interval`, component `percentile_interval` and untruncated `basic_interval`, bootstrap status, assumptions and interpretation metadata, and optional `draws`. The `reference` and `reference_interval` field names are retained for compatibility and contain the auxiliary statistic and interval.

References

Cote, M.-P., Genest, C. and Omelka, M. (2019). Rank-based inference tools for copula regression, with property and casualty insurance applications. *Insurance: Mathematics and Economics*, 89, 1–15. doi:10.1016/j.insmatheco.2019.08.001.

See Also

cjqr_eal_coef_ci, cjqr_eal_methods

Examples

```
## Not run:
dat <- cjqr_example_data(n = 200, seed = 1)
fit <- cjqr_eal(list(y1 ~ z + s(x, k = 6), y2 ~ z + s(x, k = 6)),
                data = dat, anchors = 0.5, seed = 1)
cjqr_eal_tau_ci(fit, B = 1999, seed = 1)
summary(fit, inference = "auxiliary", B = 1999, seed = 1)

## End(Not run)
```

`cjqr_example_data` *Reproducible synthetic paired data for CJQR examples*

Description

Generates a small, self-contained common-error location example with Student-t margins and a Gaussian copula. The helper generates data without fitting a model and preserves the caller's random-number state.

Usage

```
cjqr_example_data(n = 200L, seed = 1L)
```

Arguments

`n`	Integer number of independent paired observations, from 30 to 1000000. The default is 200.
`seed`	Nonnegative integer seed accepted by `cjqr_eal`; it must not exceed `.Machine$integer.max - 1000`.

Details

The continuous covariate x is uniform on $(-1, 1)$, and the binary covariate z is Bernoulli with probability 0.5. They are independent. The marginal location functions are

$$\eta_1(x, z) = 0.3 + 0.6z + 1.1\sin(\pi x),$$

$$\eta_2(x, z) = -0.2 - 0.5z + \cos(\pi x).$$

Residuals have Student-t distributions with five degrees of freedom, rescaled to standard deviations 0.8 and 0.9 respectively. A Gaussian copula induces Kendall's tau 0.5, independently of the covariates. Both conditional scales and the copula parameter are constant.

For margin j, the true conditional quantile at probability p is $\eta_j(x, z) + d_j\sqrt{3/5}\,T_5^{-1}(p)$, where $(d_1, d_2) = (0.8, 0.9)$. All quantiles therefore belong to one common response distribution, not to target-specific generating distributions.

The example is compatible with marginal formulas such as `y1 ~ z + s(x, k = 6)` and `y2 ~ z + s(x, k = 6)`. Its generating Student-t margins are not an assertion that the fitted EAL margins are exactly

Student-t. Model convergence, fit adequacy and the assumptions of any interval procedure still require assessment.

Random numbers are generated with a local, reproducible Mersenne-Twister stream. The caller's RNG kind and seed, including an originally absent seed, are restored afterwards.

Value

A data frame with n rows and four columns:

y1, y2 Paired continuous synthetic responses.

x Continuous covariate in $(-1, 1)$.

z Binary covariate coded as zero or one.

See Also

cjqr_eal, cjqr_eal_ci

Examples

```
dat <- cjqr_example_data(n = 200, seed = 1)
head(dat)

## Not run:
fit <- cjqr_eal(list(y1 ~ z + s(x, k = 6),
                     y2 ~ z + s(x, k = 6)),
                data = dat, copula = "gaussian")
summary(fit)
plot(fit, eq = 1, select = 1)

## End(Not run)
```

---

cjqr_fit_diagnostics *Inspect an EAL fit or a captured numerical convergence failure*

Description

Extracts anchor-search and initializer diagnostics without refitting, predicting, or changing the random-number stream.

Usage

```
cjqr_fit_diagnostics(object)
## S3 method for class 'cjqr_fit_diagnostics'
print(x, digits = 4L, ...)
```

Arguments

object	A fitted cjqr_eal object, or a cjqr_eal_convergence_error retained using tryCatch().
x	The object returned by cjqr_fit_diagnostics().
digits	Number of significant digits in printed numerical values.
...	Reserved; additional printing arguments are rejected.

Details

The anchor table gives initialization status, strict acceptance, objective, best known candidate objective, stationarity measure, elapsed time, and recorded issues. Initializer history includes warnings and errors even if a later retry succeeds. These are numerical diagnostics, not goodness-of-fit tests, coverage evidence, or a proof of a global optimum. A failed object remains unusable for estimates or confidence intervals.

Use `cjqr_residual_diagnostics` for marginal and joint residual displays.

Value

An object of class `cjqr_fit_diagnostics`. Its components include `accepted`, `status`, copula and fixed-df metadata, observation count, selected anchor, unresolved anchor-objective gap, an `anchors` data frame, an `initializers` data frame, selected-fit boundary flags, the numerical `certificate`, and saved runtime/seed information. Unavailable values in failed or older objects are reported as missing, not treated as successful checks.

See Also

`cjqr_eal`, `cjqr_eal_control`

Examples

```
## Not run:
d <- cjqr_example_data()
result <- tryCatch(
  cjqr_eal(list(y1 ~ z + s(x, k = 6), y2 ~ z + s(x, k = 6)),
           data = d, progress = TRUE),
  cjqr_eal_convergence_error = identity)
cjqr_fit_diagnostics(result)

## End(Not run)
```

---

cjqr_residual_diagnostics *Rosenblatt Residual Diagnostics for a Fitted Joint Model*

Description

Computes marginal and joint Rosenblatt residuals from an existing EAL model, without optimization, bootstrap, simulation or random-number generation. The plot combines two probability-scale histograms with empirical normal-score density contours.

Usage

```
cjqr_residual_diagnostics(object, newdata = NULL, y = NULL,
  order = c(1L, 2L))
## S3 method for class 'cjqr_residual_diagnostics'
print(x, digits = 4L, ...)
## S3 method for class 'cjqr_residual_diagnostics'
plot(x, breaks = 20L,
  contour_probs = seq(.1, .9, .1), grid_size = 121L,
  labels = NULL, points = TRUE, ...)
```

Arguments

`object`	A certified `cjqr_eal` fit.
`newdata`	Optional nonempty data frame of evaluation covariates. If `y` is omitted, both response columns must also be present.
`y`	Optional finite numeric matrix with one row per evaluated subject and two columns in fitted response order. Column names, if present, must match fitted response names in order. Defaults to stored analysis responses or response columns in `newdata`.
`order`	`c(1, 2)` (default) or `c(2, 1)`. The first component is marginal; the second is conditional on the first.
`x`	An object returned by `cjqr_residual_diagnostics()`.
`digits`	Number of significant digits for printed summaries, from 1 to 15.
`breaks`	Integer number of equal-width histogram bins, from 2 to 200.
`contour_probs`	One to 25 strictly increasing, distinct probability contents between zero and one for the empirical highest-density regions.
`grid_size`	Integer number of grid locations per contour axis, from 41 to 301. Increase this if the numerical density-integration check fails.
`labels`	Optional two character labels for the ordered components, for example `c("ApoB", "non-HDL-C | ApoB")`.
`points`	Logical; display every normal-score pair behind the contours.
`...`	For plotting, optional graphical controls `col`, `point_col`, `reference_col`, `cex`, `cex.axis`, `cex.lab`, `cex.main` and `lwd`. Other arguments are rejected. Printing does not accept additional arguments.

Details

For order `c(1, 2)`, write $U_j = \widehat{F}_j(Y_j \mid x)$. The ordered Rosenblatt pair is $V_1 = U_1, V_2 = \partial C(U_1, U_2)/\partial U_1$; normal scores are $W_j = \Phi^{-1}(V_j)$. The second histogram is therefore conditional, not the second marginal PIT. Reversing `order` reverses these roles. Gaussian, Student-t, Clayton, signed Frank and Gumbel copulas are supported; Student-t conditional transforms use the fitted fixed degrees of freedom plus one. Copula dependence is constant across observations.

Marginal and conditional calculations use log tails without fixed probability clipping. Numerically unrepresentable transforms fail explicitly or retain nonfinite scores with a warning; plotting never silently drops such rows. No rank normalization, whitening or centering is applied to residual scores. The plot estimates a full-covariance Gaussian kernel density with bandwidth $H = n^{-1/3}\widehat{\mathrm{Cov}}(W)$. Contour levels denote probability contents, not confidence levels. Density estimation requires at least three non-collinear finite pairs and a numerically adequate grid. The displayed axes have equal scales and retain all pairs. A horizontal uniform density-one reference is shown on the histograms; no reference contours or Q-Q plots are drawn. Graphical parameters are restored on exit.

Only retained analysis rows are returned, without padding omitted rows. Evaluation data must be complete; the function does not silently omit new rows. Row mappings are returned. All summaries and density estimates are unweighted.

The diagnostics are descriptive, not calibrated goodness-of-fit tests: fitted-parameter uncertainty, smoothing/anchor/copula selection, dependence between sampled subjects, rounding and survey design are not accounted for. Uniform histograms alone do not establish independence; small linear correlation alone does not establish joint adequacy. Fitting convergence is a separate numerical assessment.

Value

`cjqr_residual_diagnostics()` returns a `cjqr_residual_diagnostics` list containing `uniform` and `normal` ordered Rosenblatt matrices, `logF` and `logS` for their log tails, `marginal_pit` and `marginal_normal` in original response order, labels, row identifiers, copula metadata, descriptive component summaries, linear and squared-score correlations, and `joint_refits = 0L`. Printing returns this object invisibly. Plotting returns the histogram and density calculations invisibly for inspection and reproducibility.

References

Rosenblatt, M. (1952). Remarks on a multivariate transformation. *The Annals of Mathematical Statistics*, 23, 470–472. doi:10.1214/aoms/1177729394.

See Also

`residuals.cjqr_eal`, `cjqr_fit_diagnostics`

Examples

```
## Not run:
dat <- cjqr_example_data(n = 200, seed = 1)
fit <- cjqr_eal(list(y1 ~ z + s(x, k = 6), y2 ~ z + s(x, k = 6)),
                data = dat, anchors = 0.5, seed = 1)
d <- cjqr_residual_diagnostics(fit)
d$summary
plot(d)
plot(cjqr_residual_diagnostics(fit, order = c(2, 1)))
test_data <- cjqr_example_data(n = 100, seed = 2)
held_out <- cjqr_residual_diagnostics(fit, newdata = test_data)
plot(held_out, labels = c("Outcome 1", "Outcome 2 | Outcome 1"))

## End(Not run)
```

---

mgcv-smooths *Smooth-term constructors for additive model formulas*

Description

The `mgcv` smooth constructors `s`, `te`, `ti` and `t2` are re-exported for use in the two marginal formulas supplied to `cjqr_eal`. They specify bases and penalties; they do not fit the joint model themselves.

Details

See the corresponding `mgcv` documentation for arguments, basis choices, constraints and tensor-product constructions. The two margins may use different smooth terms and basis dimensions. Model support and plotting support are distinct; see `cjqr_eal_methods` for the available smooth displays.

See Also

`s`, `te`, `ti`, `t2`, `cjqr_eal`

`residuals.cjqr_eal` *Marginal diagnostic residuals for an EAL joint model*

Description

Extracts anchor-location residuals, scale-standardized residuals, marginal probability integral transforms (PITs), or marginal normal-quantile residuals without refitting or simulation.

Usage

```
## S3 method for class 'cjqr_eal'
residuals(object,
  type = c("response", "standardized", "pit", "quantile"), ...)
```

Arguments

`object`	A certified `cjqr_eal` fit.
`type`	The marginal residual definition; see Details.
`...`	Unused; unsupported arguments are rejected.

Details

Write $Y_j = \eta_j(x) + \sigma_j E_j$, with selected anchor a and $F_{E_j}(0) = a$. Response residuals are $Y_j - \eta_j(x)$; the location is the anchor quantile, not generally the conditional mean or median. Standardized residuals divide these by the fitted scale σ_j; they are not standardized to unit variance or normality.

PIT residuals are $F_{E_j}\{(Y_j - \eta_j(x))/\sigma_j\}$. Quantile residuals apply the standard normal inverse CDF to this marginal PIT. The computation uses the smaller log tail to avoid artificial infinite values from a PIT rounded to one. No probability clipping is used; genuine numerical underflow can still produce infinite normal scores.

These are marginal diagnostic quantities, not a joint Rosenblatt transform. Use `cjqr_residual_diagnostics` for conditional Rosenblatt components and the corresponding histogram/contour display. Residuals from the two margins remain dependent. Uniform/normal reference plots require adequate fitted marginal distributions; fitting and model selection are not accounted for by treating the scores as independent standard normal observations. The function is not a calibrated goodness-of-fit test.

Only the paired analysis rows are returned, in fitted order. Rows omitted by `na.action = na.omit` are not restored. For external outcomes/covariates use `predict(fit, newdata, type = "cdf", y = ...)` to obtain marginal PITs.

Value

A numeric matrix with one row per analyzed subject and two columns named by the responses. Original analysis-row names are retained when present.

See Also

`cjqr_eal`, `cjqr_fit_diagnostics`

Examples

```
## Not run:
dat <- cjqr_example_data(n = 200, seed = 1)
fit <- cjqr_eal(list(y1 ~ z + s(x, k = 6), y2 ~ z + s(x, k = 6)),
                data = dat, anchors = 0.5, seed = 1)
residuals(fit)
residuals(fit, type = "standardized")
pit <- residuals(fit, type = "pit")
z <- residuals(fit, type = "quantile")
qqnorm(z[, 1]); qqline(z[, 1])

## End(Not run)
```

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